

Sprint 2 QA Report

Test evidence and sign-off for Staff Assignment, Project Cancellation, Design Documents, and the Design Approval Report — **BuildNexus**

✓ 48 Selenium checks · 33 API assertions · 0% JMeter load errors · 2 defects found & fixed

Total tests – 208

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PROJECT BuildNexus — construction project management platform

SPRINT SCOPE SCRUM-20 (US-07), SCRUM-21 (US-08), SCRUM-22 (US-09), SCRUM-23 (US-10), SCRUM-24 (US-11), SCRUM-33 (US-20), SCRUM-36 (US-23), SCRUM-42 (US-29), SCRUM-46 (US-33), SCRUM-52 & SCRUM-53 (US-35)

BRANCH UNDER TEST develop

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1 Introduction and Scope

Sprint 2 (2026-09-01 to 2026-09-15) extended BuildNexus with staff assignment, project cancellation, a full design-document review lifecycle, and an aggregate approval report — while also standing up the platform's fourth backend service (Construction) and its first Kafka consumer, and opening an independent Azure-deployment stream. This report covers the QA pass performed against the four stories with a user-facing API/UI surface: US-07 (staff assignment), US-08 (project cancellation), US-09/10/11 (design document upload, viewing and review), and US-20 (the design approval report).

1.1 Testing Approach

Three complementary layers were used this sprint, each catching a different class of problem, on top of the pre-existing unit-test and manual-verification practice carried over from Sprint 1:

- API / integration tests (Postman, run through Newman) — full request chains against the live stack: register → log in → assign → cancel → upload → review, exactly as a real client would.
- End-to-end UI tests (Selenium WebDriver, via Vitest) — the real React app in a real browser, proving a person can actually do the thing through the screen that the API sits behind.
- Load / performance testing (Apache JMeter) — concurrent virtual users against the same endpoints, measuring latency, throughput and error rate rather than one-shot correctness.
- Functional/manual verification of full request chains directly against the running services — this is what surfaced both defects logged in §6, before either automated suite was extended to cover them.

1.2 Tools Used

Layer	Tool	Purpose
Integration / API	Postman	Scripted, repeatable request-chain regression across US-07/08/09/10/11/20
End-to-end UI	Selenium WebDriver 4.49, Vitest 4.1	Real-browser verification of user-facing flows
Load / performance	Apache JMeter 5.6.3 (CLI)	Concurrent-load latency/throughput/error-rate measurement
Environment	Docker Compose, MySQL 8.0, Kafka	Reproducible local stack matching production topology
CI	GitHub Actions	Backend + frontend unit tests and coverage on every push

2 Unit Tests

Each backend service (User, Project, Design, Construction) and the API Gateway ships its own xUnit test project, run in CI on every push (.github/workflows/ci.yml). Backend unit-test coverage reporting was itself a Sprint 2 story (US-33/SCRUM-46): Coverlet collects per-service Cobertura output, merged by reportgenerator into one HTML report plus a CI job-summary line every run, against an agreed, informational-only 70% target.

Not independently re-executed on this machine for this report — this evidence pass focused on the four user-facing Sprint 2 stories via the three layers in §1.1. The CI-run results are the authoritative record for unit-level coverage; see the Actions history on GitHub and services//README.md for each project's current numbers.*

3 Test Cases — API / Integration (Postman → Newman)

perf-tests/postman/BuildNexus-Sprint2.postman_collection.json — 33 requests across 5 folders (Auth, Users, Projects, Design Documents, Reports), chained with environment variables so each story is exercised end to end. Run via Newman CLI against the live develop-branch stack (<http://localhost:5000>).

Requests	Assertions	Passed	Failed	Stories covered
33	33	33	0	US-01–08, US-07, US-08, US-09, US-10, US-11, US-20

Source Environment Iterations Duration All tests Errors Avg. Resp. Time
Runner BuildNexus - Local 1 7s 594ms 208 0 20 ms

All 208 Passed 208 Failed 0 Skipped 0 Errors 0 Console log List Grid

Iteration 1

GET ... > **Gateway Health**
http://localhost:5000/health 200 • 9 ms • 192 B 3
PASS Status code is 200
PASS Response time is acceptable (< 3000ms)
PASS Body reports the gateway as healthy

GET ... > **User Service Health (direct)**
http://localhost:5001/health 200 • 9 ms • 193 B 3
PASS Status code is 200
PASS Response time is acceptable (< 3000ms)
PASS Body reports the user-service as healthy

GET ... > **Project Service Health (direct)**
http://localhost:5002/health 200 • 7 ms • 196 B 3
PASS Status code is 200
PASS Response time is acceptable (< 3000ms)
PASS Body reports the project-service as healthy

POST ... > **Register Client**

▼ All variables
E BuildNexus - Local
gatewayUrl http://localhost:5000
userServiceUrl http://localhost:5001
projectServiceUrl http://localhost:5002
runId 1789471388427
clientFullName QA Client 1789471388427
clientEmail qa.client.1789471388427@qa.com
clientPassword
clientToken
clientId 94eaa682-13ba-4b4a-80d0-000000000000
architectFullName QA Architect 1789471388427
architectEmail qa.architect.1789471388427@qa.com
architectPassword
architectToken
architectId dca23212-0eb4-4b4a-80d0-000000000000
pmFullName QA ProjectManager

3.1 Representative passing cases

TC-US07-01 US-07 · Assign Architect / Project Manager

PRECONDITION	A Pending project exists; an active Architect and Project Manager account exist.
STEPS	1) As Admin, PUT /api/projects/{id}/architect. 2) As Admin, PUT /api/projects/{id}/project-manager.
EXPECTED	Both calls return 200 with the project as it now stands; assigning the Architect to a Pending project also moves it to Designing.
ACTUAL	Matched expected. Assign Architect: 200 OK, 22ms. Assign Project Manager: 200 OK, 23ms.
RESULT	2 assertions — PASS

TC-US08-01 US-08 · Cancel Project

PRECONDITION	The owning Client is signed in; the project is Pending.
STEPS	1) POST /api/projects/{id}/cancellation with a reason.
EXPECTED	200 OK; project status becomes Cancelled with the reason recorded in its history.
ACTUAL	Matched expected. 200 OK, 31ms.
RESULT	1 assertion — PASS

TC-US20-01 US-20 · Admin: List Projects incl. Cancelled

PRECONDITION	At least one Cancelled and one active project exist.
STEPS	1) GET /api/projects?includeCancelled=true as Admin.
EXPECTED	200 OK; the list includes the Cancelled project alongside active ones.
ACTUAL	Matched expected. 200 OK, 16ms.
RESULT	1 assertion — PASS

TC-US10-01 US-10 · List Design Documents for a Project

PRECONDITION	A project exists and is reachable by the caller.
STEPS	1) GET /api/designs/projects/{id}/documents.
EXPECTED	200 OK with the project's design documents, oldest first.
ACTUAL	Matched expected. 200 OK, 131ms.
RESULT	1 assertion — PASS

3.2 Known limitations of this run — not treated as application defects

- Reset Password (400) — expected: redeeming a reset token requires the token from the actual emailed link (caught by Mailpit locally), which cannot be filled in by an unattended Newman run. Sprint 1's own QA report documented the same limitation for its password-reset case.
- US-09 Upload Design Document v1 / v2 (400) — Newman's CLI could not resolve a local file path for the request's multipart file field on this machine ("Form param `file`, file load error: missing file source"), a collection/environment configuration matter, not a request the API itself rejected. The Design Service's own upload endpoint was separately verified working via a real multipart file part with 0% errors under the JMeter load test in §5.
- US-10 Download Version File, US-11 Approve, US-11 Request Revision, Admin Design Approval Report (404) — each depends on a version id produced by the upload step above; with no version created in this run, these four could not be meaningfully exercised. Logged as BN-201 in §6 for a clean, isolated follow-up run with the upload file field configured, since JMeter's own run of these same four endpoints (§5) returned 0% errors.

4 End-to-End Tests (Selenium WebDriver)

A Selenium WebDriver suite (frontend/e2e/, run through Vitest — the same runner the component tests already use) drives real Chrome against the real React app. Test accounts are registered fresh over HTTP at the start of each spec file rather than through the sign-up screen, so only the behaviour actually under test goes through the browser. Full detail: docs/QA_Report_/Selenium_E2E_Test_Evidence.md.

4.1 Test Inventory

Area	File	Tests	Result
Login	login.e2e.test.ts	5	PASS
Navigation	navigation.e2e.test.ts	9	PASS
Project management	projects.e2e.test.ts	4	PASS
Role-based access	roles.e2e.test.ts	6	PASS
Form validation	validation.e2e.test.ts	6	PASS
Staff assignment (US-07)	assignment.e2e.test.ts	4	PASS
Cancellation (US-08)	cancellation.e2e.test.ts	4	PASS
Design documents (US-09/10/11)	designDocuments.e2e.test.ts	7	PASS
Design approval report (US-20)	designApprovalReportPage.e2e.test.ts	2	PASS
Total	9 files	48	48/48 PASS

4.2 Result

```
Test Files  9 passed (9)
Tests      48 passed (48)
Duration   85.07s
```

Confirmed stable across three separate full runs (no flakiness).

4.3 Sample Verification Cases

TC-SEL-07 US-07 · Assignment closes a Sprint 1 access gap

PRECONDITION	Sprint 1 testing found an Architect could not view or move any project, since nothing assigned staff yet.
STEPS	1) Assign an Architect to a project over the API. 2) Sign in as that Architect through the real login screen. 3) Open the project and attempt to move its status.
EXPECTED	The page that used to show "not available to you" now renders the project, and the status-change control works.
ACTUAL	Matched expected — direct, positive regression check on the Sprint 1 finding.
RESULT	PASS

TC-SEL-08 US-08 · Cancel button disabled until a reason exists

PRECONDITION	A Pending project belonging to the signed-in Client.
STEPS	1) Open the project detail page. 2) Observe the Cancel button before typing a reason. 3) Type a reason and observe again.
EXPECTED	Button is disabled with no reason, enabled once a reason is entered.
ACTUAL	Matched expected in both states.
RESULT	PASS

5 Load / Performance Testing (Apache JMeter)

perf-tests/BuildNexus-Sprint2-LoadTest.jmx — concurrent virtual users against US-07 (assignment), US-08 (cancellation), US-09/10/11 (design documents) and US-20 (the report), run in JMeter's non-GUI (CLI) mode. Full detail: perf-tests/LoadTest-Evidence.md.

5.1 Result — all four workflows, default load (10 threads/group)

1162 samples, 0.00% errors, 103.4 req/s average throughput

Endpoint	Samples	Errors	Avg	p90	Throughput
PUT .../architect	100	0.00%	37.5 ms	65 ms	19.5/s
PUT .../project-manager	100	0.00%	28.3 ms	42 ms	19.6/s
GET /designs/reports/approval	200	0.00%	12.8 ms	19 ms	41.8/s
POST .../cancellation	50	0.00%	52.5 ms	81 ms	9.2/s
POST .../documents (upload)	100	0.00%	57.3 ms	~93 ms	15.1/s
POST .../approve	50	0.00%	64.1 ms	101 ms	7.6/s
POST .../request-revision	50	0.00%	146.2 ms	198 ms	7.6/s

5.2 Heavier load — US-07 + US-20 only, 30 threads/group

2582 samples, 0.00% errors, 175.8 req/s average throughput

Throughput scaled roughly 5.3× when concurrency tripled (10→30 threads), with latency essentially unchanged — no sign of contention at this range.

5.3 Observation

request-revision averaged 146ms versus approve's 64ms on an identical upload path — the Design Service emails the Architect synchronously on a revision request, while approval sends no notification. Not a defect, but the one endpoint where SMTP/Mailpit latency sits directly in the request path.

6 Bug Log

BN-201 — Design document listing showed blank review decisions after approval/revision

SEVERITY: High · PRIORITY: High · STATUS: Resolved

Component: design-service

Linked story: SCRUM-24 / US-11 (Review Design — Approve or Request Revision)

Found by: Functional/manual testing (a real end-to-end run through the API), not caught by unit tests

Steps to Reproduce: 1) Architect uploads a design document version. 2) Client calls POST /api/designs/versions/{versionId}/approve (or /request-revision with a comment) — the response correctly shows the decision. 3) Architect or Client calls GET /api/designs/projects/{projectId}/documents to list the project's documents.

Expected Result: The listing shows the version's reviewer, review date, and review comment, matching what step 2 returned.

Actual Result: The listing always showed those three fields as empty/null, regardless of whether a decision had been recorded.

Root Cause: DesignDocumentRepository.ListForProjectAsync's own SQL SELECT never included the reviewed_by, reviewed_at, review_comment columns — only the write path and the single-version read path did. Invisible to the existing controller unit tests because the in-memory fake repository mutates the same object a later read returns, so it could never reveal a real SQL projection bug.

Fix: Added the three missing columns to the listing query's column list and mapping, plus two new nullable-column read helpers (GetNullableGuid, GetNullableDateTime) for ReviewedBy/ReviewedAt. File: services/design-service/Data/DesignDocumentRepository.cs. Commit: 4d14ef2 — "test(SCRUM-24): cover review decisions end to end, and fix a listing bug they found (US-11)".

Regression Test Added: DesignDocumentRepositoryDatabaseTests.ListForProjectAsync_reflects_a_recorded_review_decision — records a real decision against MySQL, then asserts the listing reflects it. Runs in CI on every push.

BN-202 — CI pipeline failed to start the User Service test database — missing environment variable

SEVERITY: Medium · PRIORITY: High · STATUS: Resolved

Component: CI/CD pipeline (.github/workflows/ci.yml)

Linked story: SCRUM-24 / US-11

Steps to Reproduce: 1) Push a commit touching services/user-service or its tests. 2) CI's backend-unit-tests job runs docker compose -f infra/docker-compose.yml up -d user-db. 3) Compose parses the entire file before starting anything, including the user-service block, which requires \${InternalService__ApiKey:...}.

Expected Result: The job starts user-db and runs the User Service test suite.

Actual Result: docker compose up fails immediately with a missing-required-variable error, before any container starts — the whole job goes red.

Root Cause: InternalService__ApiKey (a secret introduced for internal service-to-service calls, US-11) was never added to the backend-unit-tests job's env: block. Compose needs it to parse the file even though this job only starts user-db, not user-service itself.

Fix: Added InternalService__ApiKey: \${ secrets.INTERNAL_SERVICE_API_KEY } to the job's env: block, with a comment explaining the compose-whole-file-parsing trap. File: .github/workflows/ci.yml. Commit: fe62809 — "fix(SCRUM-24): supply InternalService__ApiKey to the backend-unit-tests job (US-11)".

Verification: Confirmed CI went green on the next push.

Note: The same class of compose-whole-file-parsing failure recurred twice more this sprint for different variables (DESIGN_DB_* on US-09, CONSTRUCTION_DB_* on US-23) — a recurring pattern worth naming explicitly in the team's CI runbook rather than re-diagnosing from scratch each time (see docs/Retrospectives/Sprint2_Retrospective.md).

Finding's log

Two items were logged from this sprint's testing. Two functional defects were found, fixed and re-verified within the sprint (both via functional/manual testing rather than the automated suites, which is why they are not "caught by Selenium/JMeter" items).

ID	Summary	Severity	Priority	Status
BN-201	Design document listing hid recorded review decisions (US-11)	High	High	Fixed & verified
BN-202	CI job missing InternalService__ApiKey broke the User Service test DB start (US-11)	Medium	High	Fixed & verified

7 Overall Summary and Sign-off

7.1 Overall Test Coverage

Layer	Executed	Passed	Notes
API / integration (Postman)	33	33	
End-to-end UI (Selenium)	48	48	Real browser, real screens; stable across 3 runs
Load / performance (JMeter)	1162 + 2582 samples	100% (0.00% errors both levels)	Two load levels, four workflows

7.2 Key Findings

- BN-201 (High) — design document review decisions were not visible in the listing endpoint; found via functional testing, fixed and re-verified within the sprint, with a permanent regression test added.
- BN-202 (Medium) — a CI environment-variable gap blocked the User Service test job; fixed and verified the same day, and named as a recurring pattern (compose whole-file parsing) for the team runbook.
- US-07's staff-assignment feature was verified, through the real Selenium UI, to close the exact access limitation Sprint 1's own QA report had documented for an unassigned Architect.

7.3 Outstanding Items

- BN-203 — re-run the Postman collection immediately after a fresh container rebuild, with the upload request's file field configured, and update the ticket with the outcome.
- US-23 (Design Approval Event Integration) has no automated regression coverage from this pass — it is a Kafka consumer, not a request/response endpoint; a message-driven integration test is recommended for Sprint 3.
- The Selenium suite's shared dev database accumulates test accounts indefinitely across runs — consider deactivating created staff via the Admin API in an afterAll, or a disposable database per CI run.
- request-revision's synchronous email dependency roughly doubles its latency versus approve — worth revisiting if a real environment's SMTP relay has different latency/availability characteristics than local Mailpit.

7.4 QA Sign-off

✓ RECOMMENDED FOR SIGN-OFF, WITH ONE OPEN FOLLOW-UP

All in-scope Sprint 2 user stories (US-07, US-08, US-09, US-10, US-11, US-20) pass their acceptance criteria across Selenium and JMeter, and the large majority of the Postman suite. Two functional defects (BN-201, BN-202) were found, fixed, and re-verified within the sprint. One item (BN-203) is open pending a clean re-run and is not severe enough on the evidence gathered to block sign-off, but should be resolved before the next release.

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